

Sri Datta Ganesh Bandreddi

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EDUCATION

Carnegie Mellon University

M.S. in AI Engineering (Biomedical Engineering) | GPA: 4.00/4.00

Pittsburgh, PA
Expected Dec 2026

- **Coursework:** Introduction to Deep Learning (24-788), Learning for 3D Vision, AI and IoT.
- **Teaching Assistant** — “AI and IoT”; built teaching material including an IoT VLM Colab notebook.

Visvesvaraya National Institute of Technology (VNIT)

B.Tech, Mechanical Engineering | GPA: 8.44/10.00 (Scholaro-scaled 3.63/4.00)

Nagpur, India
Jul 2022

- **Coursework:** AI in Manufacturing, Industrial Robotics, Manufacturing Process Automation, Computer Graphics & Solid Modelling, Data Structures, Finite Element Method, Biomechanics, Machine Design.

SKILLS

Languages: Python, C++, Go, Java, SQL, Bash / Shell scripting

GPU / Acceleration: CUDA, cuDNN, cuBLAS, TensorRT, Triton Inference Server, NCCL, Nsight Systems/Compute

LLM / Agents: LangGraph, LangChain, RAG, agentic tool-calling, FastAPI (serving)

DL Frameworks: PyTorch, JAX, TensorFlow, Hugging Face, NeMo / Megatron-LM, LoRA / QLoRA, FP8 / BF16 mixed precision

Data & Big Data: RAPIDS (cuDF, cuML), NVTabular, DALI, NumPy, SciPy, Pandas, SparkML, Ray, Apache Spark, H3, medallion architecture

Systems & MLOps / IaC: Docker, Kubernetes, Slurm, Terraform, Bazel, Jenkins / GitHub Actions, Helm, MLflow, Git, SonarQube

Cloud: AWS (ECS, SageMaker, S3, RDS, KMS, EventBridge), GCP

Databases: PostgreSQL, Citus, Neo4j, memgraph, JanusGraph, MongoDB, Cassandra, ScyllaDB, Redis, Valkey, Kafka, Iceberg, Elasticsearch, FAISS

Security & Observability: Prometheus, OpenReplay, AWS KMS / encryption-at-rest, BYOK, GDPR, SAML / SSO / RBAC / OAuth

Frontend: ReactJS, Next.js

PROFESSIONAL EXPERIENCE

Motional

Autonomous-vehicle data platform

Data Platform Intern (Infrastructure)

May 2026 – Aug 2026

- Designed, prototyped, and deployed a metadata-extraction system that characterizes AV drivetlogs before full extraction, enabling teams to query ~1 TB drivetlogs (petabyte-scale corpus) without processing the raw logs — delivered a staging PoC and authored the production design.
- Architected the production system as a **medallion architecture** to stand it up as a reusable, future-facing data product.
- Enriched drivetlog data with weather and map layers and used **H3 geospatial indexing** to compute area-based aggregation and rollout metrics.
- Built novel indexing for geospatial datasets enabling fast retrieval on scene metadata, and indexed scenes via embeddings for similarity search — **filed a provisional patent** for the approach.
- Augmented the platform with an **agentic AI pipeline (LangGraph)** for natural-language drivetlog retrieval via tool calls, served through a **FastAPI** backend.
- Benchmarked **Ray Data pipelines vs. Spark** on Kubernetes; final design used Ray with AWS EventBridge, S3, and RDS.

Phenom

Hyderabad, India

Product Development Engineer, Enterprise Talent Graph

Dec 2023 – Jul 2025

- Led a team of six engineers to re-architect a multi-tenant AI analytics platform, enhancing scalability and reliability under concurrent tenant load by refactoring shared services and reducing cross-tenant coupling.
- Scaled a production knowledge graph to 1M+ entities with continuous ingestion, by designing validated aggregation pipelines integrated with Neo4j graph schemas and Iceberg versioned storage.
- Ensured high availability for workloads serving 100+ enterprise clients, by managing distributed Kafka, PostgreSQL, and Redis clusters and building CI/CD pipelines using Jenkins and Helm charts on Kubernetes.
- Accelerated AI-driven query response times by 40% across 100+ clients, by tuning PostgreSQL horizontal sharding strategies.
- Owned the full Kafka platform: built a custom Kafka exporter (shell scripting + Prometheus scraping) and Kafka UI for cluster management, giving the team live visibility into partitioning, rebalancing, consumer-lag, and node failures; led the migration of Kafka from ZooKeeper to **KRaft**.
- Pioneered running Redis and other databases on Kubernetes for elastic scaling, and owned the Helm charts that made those deployments repeatable.
- Member of the **central crypto team**: implemented encryption at rest and owned key management/rotation via AWS KMS; helped build **Bring-Your-Own-Key (BYOK)** support for EU customers.
- Enforced **GDPR data-retention** policies across data products, managing PII safety and guaranteed deletion.

- Redesigned and rewrote a **proxy service in Go**, reducing startup time and **increasing throughput by 30%**; implemented Docker Compose integration-testing pipelines.
- Benchmarked datastores (ScyllaDB, Cassandra, PostgreSQL, Citus) and built RAG pipelines over related graph entities; deployed and maintained OpenReplay for UI-behavior heatmaps and product insights; negotiated long-term Neo4j licensing.
- *Recognition*: Phenomenal Award for Outstanding Performance (both Q1 and Q2, 2024).

BYJU'S

Hyderabad, India

Software Engineer, Core Platform Team

Jul 2022 – Dec 2023

- Cut infrastructure costs by 70% while maintaining reliability for 1M+ users, by deploying scheduled AWS ECS auto-scaling policies (Terraform-authored IaC) based on user traffic patterns.
- Eliminated dependency on external identity services by 50%, by leading development of an in-house authentication and authorization platform using Go and Java (Spring Boot).
- Led an internal **SSO** project end-to-end — implementing SAML, SSO, and RBAC so every inter-service call was authenticated, authorized, and traceable — while mentoring and leading interns.
- Built a **multi-channel communications platform** with real-time targeting that tracked exact communication volume, enabling reconciliation that **saved at least \$100K** and **boosted user engagement by 20%**.
- Ran load testing on target APIs that **cut latency by 30%**; engineered an in-house CRM backend in Java (Spring Boot), delivered end-to-end within two months.
- Held admin over 5 repositories as a fresher and owned the full auth + comms flow; survived 7 rounds of layoffs through the company's decline, a signal of critical ownership.

Phenom

Hyderabad, India

Machine Learning Intern

May 2021 – Jun 2022

- Boosted recommendation accuracy by building persona classification models using NLP (spaCy, BERT embeddings) across 3 data sources, serving personalized content for production recommendation pipelines.
- Accelerated recommender feature preprocessing by migrating CPU pipelines to NVTabular on GPU, reducing latency and validating scalability across 2 production databases.

PUBLICATIONS & RESEARCH

- **FLARE: Fast Low-rank Attention Routing Engine** — V. Puri, A. Joglekar, **S. D. G. Bandreddi**, K. Ferguson, Y. Chen, Y. J. Zhang, L. B. Kara. arXiv:2508.12594; submitted to TMLR (*3rd author*). Later adopted by an NVIDIA × General Motors crash-dynamics effort (arXiv:2605.27758) via a FLARE-based GeoTransolver backbone on NVIDIA PhysicsNeMo (~2× memory reduction).
- **FLARE++: Low-rank Attention with Dynamic Attention Routing** — arXiv:2608.11519, ongoing, targeting ICLR 2026 (*2nd author*). Improves latent-token routing over FLARE.
- **Comparative FEA Study of Palatal Expansion Devices** — **S. D. Bandreddi**, V. Vaidya, R. R. Salguti, R. Uddanwadiker. Presented at ICMAE 2022, Bratislava, Slovakia (*first author*).

SELECTED PROJECTS

FLARE / Efficient Attention (CMU Research)

Aug 2025 – Present

- Validated >200× speedups and sub-quadratic memory scaling for linear attention on Long Range Arena (1k–8k tokens); profiled multi-GPU NCCL training across FP8/BF16 with Nsight Systems on Slurm.
- Owned the LRA benchmark suite and latent-expression ablations; extended FLARE to image classification and diffusion, including all architecture changes and GPU image data pipelines (NVIDIA DALI).

FLARE++, Dynamic Attention Routing (CMU Research)

Ongoing

- Ran token-mixer ablations (AB-UPT vs. FLARE) on DrivAerML and PDEBench to evaluate dynamic attention routing and isolate which design components drive accuracy/efficiency gains.

CT-Transformer, Vision Transformers (CMU)

Sep 2025 – Dec 2025

- Achieved 96.22% macro AUC-ROC on 15-class VinBigData chest X-ray classification, outperforming ResNet-50, EfficientNet-B3, and ViT baselines, with a 210M-parameter Swin Transformer and fused attention kernels.
- Optimized inference latency 3× over PyTorch eager by converting to TensorRT (FP16/INT8), accelerating preprocessing with DALI, and deploying on Triton with adaptive batching.

Distributed LLM Fine-Tuning with Megatron-LM (CMU)

Aug 2025 – Oct 2025

- Fine-tuned a 7B-parameter LLM with stable convergence and ~40% reduced memory on 4× H100 GPUs, via tensor/pipeline parallelism (NCCL), gradient checkpointing, and LoRA/QLoRA adapters.
- Achieved 2.1× training throughput over naive data parallelism by tuning NCCL, gradient bucketing, and optimizer-state sharding.

EMG Hand Gesture Recognition (CMU)

Sep 2025 – Oct 2025

- Built PyTorch LSTM/CNN + attention models for EMG hand-gesture classification, achieving 85–92% accuracy on 16-channel, 1000 Hz signals, with controlled ablations over subject-wise splits.

GitAsk, Local-First RAG over GitHub Repos (Open Source)

- Core contributor to a browser-based tool that turns any repo into a conversational assistant with an in-browser LLM (no backend). Implemented **BM25** sparse retrieval for hybrid search, embeddings (MiniLM), security hardening (encrypted key vault), and performance optimization.

AI-Driven Knowledge Graph for Obsidian (Personal)

Jul 2024 – Present

- Auto-links notes by extracting entities/relationships with spaCy and BERT, with real-time graph clustering (NetworkX) and D3.js visualization.

Go Lopper (URL Shortener) & Trello Clone (Personal)

- Go Lopper: URL shortener in Go using ULID keys in PostgreSQL for scalable inserts, Redis caching, Docker-deployable.
- Trello Clone: Next.js + Appwrite to-do app with drag-and-drop, GPT-API task summaries, and Google auth.

Comparative FEA of Palatal Expansion Devices (B.Tech Capstone)

2022

- Conducted FEA comparing MARPE and HYRAX devices, modeling skull stress/displacement with suture lines; showed MARPE's superior stress reduction. Accepted for presentation at ICMAE 2022 (first author).

AWARDS & HONORS

- **Phenomenal Award for Outstanding Performance** — Phenom, Hyderabad (2024), for exceptional contributions in Q1 and Q2.
- **Perfect GPA** — 4.00/4.00 at Carnegie Mellon University.
- **TartanHacks track winner** (early 2026) — built a playable game representing CMU in under 15 KB total.
- **Provisional patent** (Motional, 2026) — geospatial indexing / retrieval for AV scene metadata.

LEADERSHIP & EXTRACURRICULAR

- **Student Mentor**, VNIT (2020–2022) — mentored ~120 first-year Mechanical Engineering students.
- **Student Council Representative**, VNIT (2018–2022) — represented the student body to faculty and improved course plans.
- Organizing Committee, Aarohi (VNIT cultural fest, 2018); active member of the Cycling and Quiz clubs; district-level badminton (2019).
- **Languages:** English (Proficient), Hindi (Fluent), Telugu (Native), German (Basic).